Félix Laplante

Personal Information

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Education

2024–2026	**Master in Mathematics and AI, Orsay** **Ranked 1st, 19.28/20 in M1, 18.28 in M2** Joint coursework with the Probability and Statistics program, co-organized with Institut Polytechnique de Paris.
2021–2024	**Bachelor in Mathematics and Applications, Évry** **Ranked 1st, 19.425/20 in L3**

Research experience

April–September 2026	**Research internship in Causality under Christophe Ambroise and Pierre Humbert, LaMME** Study of identifiability in structural causal models, with a focus on non-Gaussian settings and causal discovery. Analysis of quantum causal inflation constraints and their implications for classical causal representations.
April–September 2025	**Research internship on Joint Models under Estelle Kuhn, Christophe Ambroise, INRAE** Development of estimation and prediction methods for joint models and extension to multi-state models. Derivation of uniform high-probability bounds for survival functions, implementation of the `jmstate` package, simulation studies and analysis on real-world data, and comparison with state-of-the-art methods.
April–June 2024	**Research internship in Clustering under Christophe Ambroise, LaMME** Design and implementation of a hierarchical topological clustering algorithm inspired by SVMs, *Spectral Bridges*. Partitioning into Voronoi regions, followed by grouping and automatic detection of boundaries and low-density regions using spectral methods on a local scale-invariant affinity matrix.

Awards / Grants

June 2024	Sophie Germain Excellence Scholarship

Published papers / Preprints

May 2026	**A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores** *Félix Laplante* arXiv preprint https://arxiv.org/abs/2605.25852.
October 2025	**A General Framework for Joint Multi-State Models** *Félix Laplante, Christophe Ambroise* arXiv preprint https://arxiv.org/pdf/2510.07128 submitted to *Statistics and Computing*.
December 2024	**Scalable Spectral Clustering Based on Vector Quantization** *Félix Laplante, Christophe Ambroise* Paper published in *Computo* https://doi.org/10.57750/1gr8-bk61.

Talks / Posters

June 2026	**A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores**

	Poster presentation at Ukrainian Mathematics Days (IHP) discussing the PIT-CP framework for conditional validity based on pivotal scores.
June 2026	**A General Framework for Joint Multi-State Models** Oral presentation at *Journées de la Statistique* (Clermont-Ferrand) introducing a unified framework for longitudinal biomarkers and multi-state event processes.
September 2025	**Uniform High-Probability Bands for Survival Functions** Poster presentation at *StatMathAppli* (Fréjus) deriving asymptotic uniform confidence bands under a joint modeling framework.
June 2025	**Selecting the Number of Clusters in a Clustering Algorithm** Oral presentation at *Journées de la Statistique* (Marseille) introducing a connectivity-based score for scalable spectral clustering.

Other attended events

March 2026	Workshop on Uncertainty Quantification and Causality
October 2025	Winter School on Causality and Explainable AI
July 2025	Working group on Topological Data Analysis (LaMME)
June 2025	Working group on Causality (MIA Paris-Saclay)

Services

May 2026	Co-organizer of a Causality Working Group with Pierre Humbert

Packages

Python

`jmstate`
Companion package to the *Multi-State Models* paper, built on `PyTorch`.

`pitcp`
Companion package to the *Pivotal Scores Conformal Prediction* paper, built on `zuko`.

`sbcluster`
Companion package to the *Scalable Spectral Clustering* paper.

`fastkmeanspp`
C++-backed implementation of K-means++, with massive speedups over `scikit-learn`.

`mimisbm`
Vectorized port of the corresponding R project, implementing a mixture of multilayer stochastic block models.

`uniformbands`
Asymptotic uniform confidence bands for survival functions.

Languages

French	Native
English	Native

Interests & Activities

Hobbies	Photography, cycling, piano, hiking